

Rakshit Pradhan

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EDUCATION

Purdue University
Bachelor of Science in Physics

- Semester Honors

West Lafayette, IN

TECHNICAL SKILLS

Languages: Python, C, JavaScript, HTML/CSS
ML / Data: PyTorch, Scikit-learn, Pandas, NumPy, TensorBoard
Web / Tools: React.js, Node.js, Express.js, SQL, MongoDB, Metabase, CVAT

WORK EXPERIENCE

Undergraduate Machine Learning Researcher August 2025 - Present
IDEAS Lab - Purdue University

- Trained and evaluated **MultiMAE- and GR-ConvNet-based vision models** in Python/PyTorch on the Jacquard dataset (52K images) using RGB, depth, and segmentation modalities
- Integrated a pretrained **MultiMAE encoder with GR-ConvNet and SE-ResUNet decoders**, experimenting with multimodal feature fusion for **robotic grasp prediction**
- Applied **LoRA-based fine-tuning to custom encoder-decoder grasping models** to improve training efficiency and overall prediction performance
- Tracked loss curves and IoU in TensorBoard to compare architectures and guide iterative improvements, **increasing performance from 10% to 52% IoU**

Undergraduate Researcher August 2025 - December 2025
Purdue Vertically Integrated Projects, Advisor: Amy Reibman

- Developed and evaluated **computer vision pipelines in Python/PyTorch** on cow video data for behavior analysis and livestock monitoring
- Designed a **reference-motion evaluation method using sparse keypoints and monocular video**, reducing dependence on expensive multi-camera setups
- Annotated cow keypoints in CVAT** and contributed to dataset curation and preprocessing, improving motion-estimation model performance by **20–25% through higher-quality labels**
- Project report: [GitHub link](#)

Growth Engineering AI Intern May 2025 - July 2025
Wondercraft (YC S'22)

- Built analytics dashboards in **SQL, Python, and Metabase to track product, growth, and subscription metrics**, helping leadership turn raw usage data into clearer operational insights
- Developed internal **Python and React tools** to monitor user-growth trends and support faster, more structured analysis of funnel behavior and conversion signals
- Improved **data pipelines and debugging workflows** for growth analytics, increasing data reliability and enabling more consistent experimentation across product and marketing efforts

Undergraduate Research Fellow Jan 2025 - May 2025
Purdue Industrial Engineering

- Investigated full-body joint kinematics during lifting tasks using **IMU-based motion capture systems and computer vision**, generating structured datasets for ergonomic and biomechanical analysis
- Cleaned, synchronized, and **analyzed motion-capture data in Python**, extracting three-axis acceleration, velocity, and magnitude features from joints across the pelvis, spine, neck, shoulders, arms, hands, and lower body to **study load-dependent movement patterns**
- Analyzed relationships between joint kinematics and physical load and tested **machine learning models such as Random Forest and XGBoost** to classify or predict lifting weight from motion data, supporting the development of faster, non-intrusive ergonomic assessment methods

Machine Learning Engineering Intern May 2023 – August 2023
PerspectAI, T-Hub

- Assisted in the development of **MirrorAI**, an AI-driven tool for personalized career insights, applying skills in **prompt engineering and model optimization** to help innovate product features and functionalities
- Administered data preprocessing initiatives to improve the accuracy of Machine Learning models, achieving a **15-20% improvement** in overall model performance
- Researched and successfully implemented cutting-edge **machine learning techniques** from recent academic papers and GitHub repositories

PROJECTS AND CONFERENCES

DriftDreamer | *Python, PyTorch, Gazebo, World Models* January 2026 - Present

- Built a **world-model-based navigation system in Gazebo for warehouse robots**, using a **PyTorch RSSM and CEM-MPC** to control a simulated tugbot along routes with obstacle-aware planning
- Developed an **end-to-end pipeline for collecting robot trajectory data, processing lidar inputs**, training the world model, and testing it in closed-loop navigation
- Improved **long-horizon prediction and planning reliability** by redesigning the training setup around full trajectories, richer sensor inputs, and iterative model refinement
- Project Link: [GitHub link](#)

www.senca.net | *React.js, JavaScript, Python, Node.js, Express.js, MongoDB, HTML, CSS, Git* November 2024 - Present

- Designed and engineered the front-end of **www.senca.net**, an AI content aggregator that curates and displays relevant AI resources for users, utilizing **React.js, JavaScript, HTML, CSS, and Figma** to create an interactive and user-friendly interface
- Architected and implemented the website's backend using **Node.js, Express.js, and MongoDB**, achieving a robust, scalable, and secure system that ensures smooth website performance
- Developed algorithms using Python** for web scraping to automate data gathering, resulting in significantly improved efficiency and a substantial reduction in manual effort and time

Conferences

- Purdue Undergraduate Fall Expo 2025
- Purdue Undergraduate Spring Expo 2025